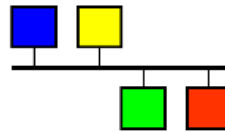


DE LA RECHERCHE À L'INDUSTRIE



EPICS TOOLS FOR SMALL



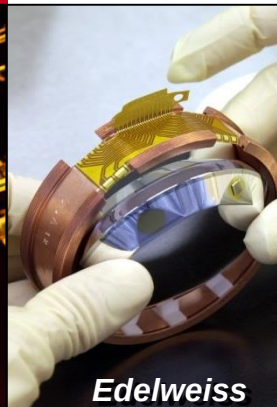
EXPERIMENT BASED ON PLC



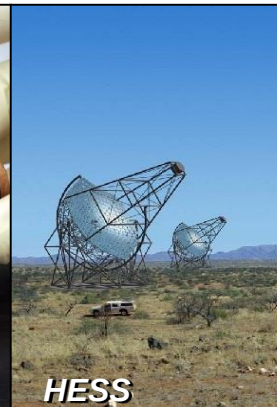
Double Chooz



ALICE



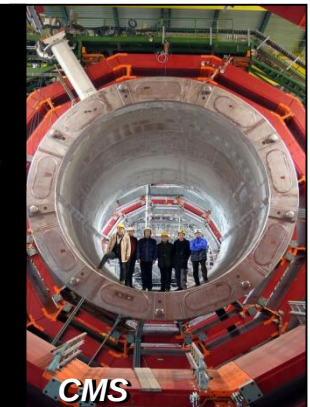
Edelweiss



HESS



Herschel

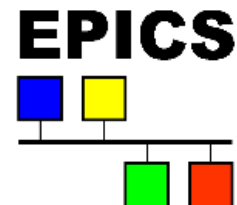


CMS



Katy SAINTIN & Paul LOTRUS

June 2019



Small Experiments

- In-house projects
- **2 developers**

Functions :

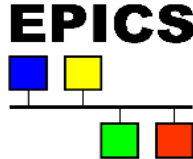
- Windows OS
- SIEMENS PLC Protocol
- Optimized archiving
- Tool to build server
- Tool to build GUI
- Web remote access

Big facilities :

- International projects
- **EPICS community**

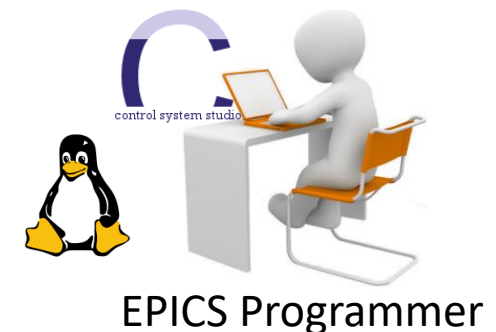
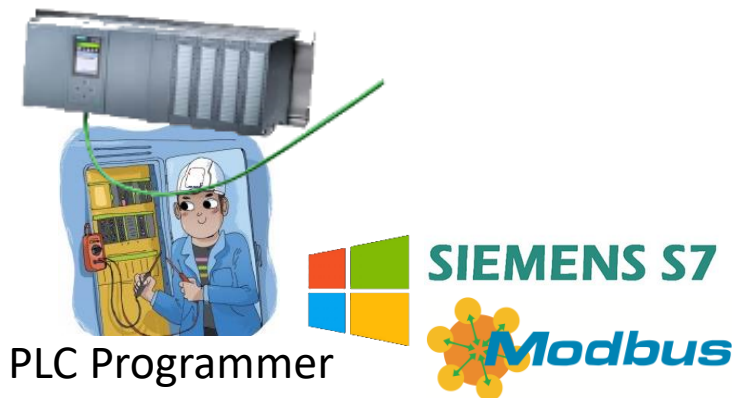
Functions :

- Linux OS
- **Numerous Drivers**
- **Archiving Appliance**
- **CSS BOY to build a GUI**

- **Plcparsertool**   
- **CAFE Java** (Channel Access For EPICS) 
- **DXF2OPI**   
- **MOONARCH** (Memory Optimizer On ARCHiver Appliance)
- **EPICS WEB**  
- **DATABROWSER** 

Generate all EPICS files for build an EPICS IOC

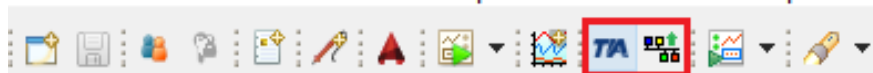
- db file
- include.dbd file
- cmd file for protocol configuration



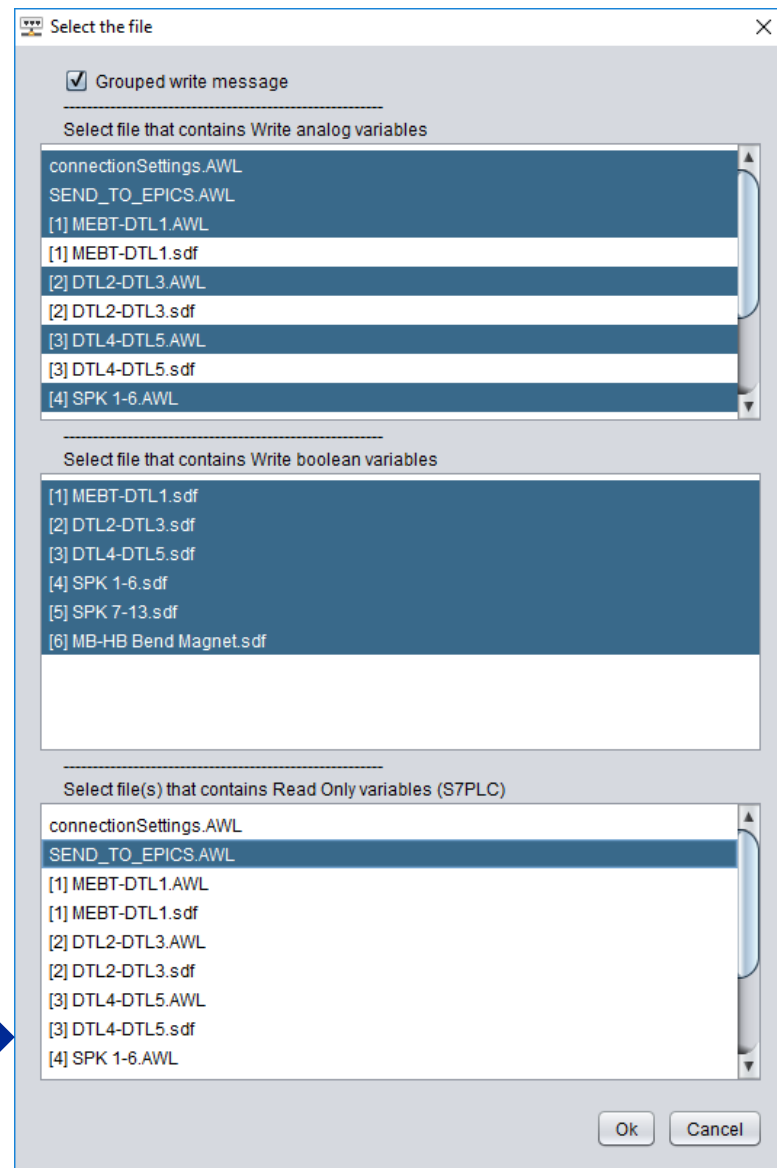
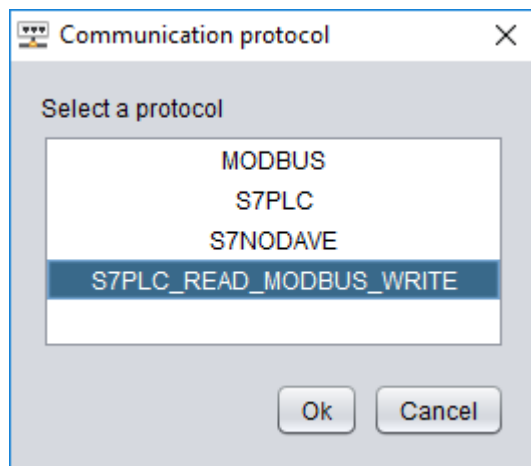
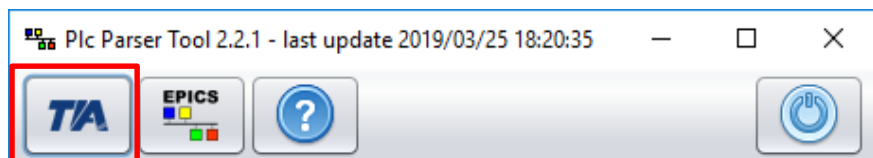
CSS plugin

CS-Studio

File Edit Search CS-Studio Sample Menu Window Help



Standalone



Ess_nblm

Search

Index

Index	Name	Record type	Address	Data type	DTYP
358	FEB-050Row:PBI-PLC-Line6:HardErr	hi	@PLC01/29 T-BYTE B-A	BYTE	S7nls
359	Ess_nblm				
360					
361	Search				
362					
363	Index	Name	Record type	Address	Data type
364	126	FEB-050Row:PBI-PLC-Line4:ShuntErrTim	ao	@asyn(PLC01505_AS_WRITE_0,44,1)INT16	INT16
365	34	FEB-050Row:PBI-PLC-Line1:PrsLoErrTrsh	ao	@asyn(PLC01502_AS_WRITE_0,45,1)FLOAT32_BE	FLOAT32_BE
366	65	FEB-050Row:PBI-PLC-Line2:PrsLoErrTrsh	ao	@asyn(PLC01503_AS_WRITE_0,45,1)FLOAT32_BE	FLOAT32_BE
367	96	FEB-050Row:PBI-PLC-Line3:PrsLoErrTrsh	ao	@asyn(PLC01504_AS_WRITE_0,45,1)FLOAT32_BE	FLOAT32_BE
368	127	FEB-050Row:PBI-PLC-Line4:PrsLoErrTrsh	ao	@asyn(PLC01505_AS_WRITE_0,45,1)FLOAT32_BE	FLOAT32_BE
369	35	FEB-050Row:PBI-PLC-Line1:PrsLoErrTim	ao	@asyn(PLC01502_AS_WRITE_0,47,1)INT16	INT16
130	66	FEB-050Row:PBI-PLC-Line2:PrsLoErrTim	ao	@asyn(PLC01503_AS_WRITE_0,47,1)INT16	INT16
131	97	FEB-050Row:PBI-PLC-Line3:PrsLoErrTim	ao	@asyn(PLC01504_AS_WRITE_0,47,1)INT16	INT16
132	128	FEB-050Row:PBI-PLC-Line4:PrsLoErrTim	ao	@asyn(PLC01505_AS_WRITE_0,47,1)INT16	INT16
133	36	FEB-050Row:PBI-PLC-Line1:GapFlw	ao	@asyn(PLC01502_AS_WRITE_0,48,1)FLOAT32_BE	FLOAT32_BE
134	67	FEB-050Row:PBI-PLC-Line2:GapFlw	ao	@asyn(PLC01503_AS_WRITE_0,48,1)FLOAT32_BE	FLOAT32_BE
135	98	FEB-050Row:PBI-PLC-Line3:GapFlw	ao	@asyn(PLC01504_AS_WRITE_0,48,1)FLOAT32_BE	FLOAT32_BE
136	129	FEB-050Row:PBI-PLC-Line4:GapFlw	ao	@asyn(PLC01505_AS_WRITE_0,48,1)FLOAT32_BE	FLOAT32_BE
137	6	FEB-050Row:PBI-FT-A10:FlwR	ai	@PLC01/30 T=FLOAT	FLOAT
138	7	FEB-050Row:PBI-FT-A19:FlwR	ai	@PLC01/34 T=FLOAT	FLOAT
139	8	FEB-050Row:PBI-FCV-A10:PosSPR	ai	@PLC01/38 T=FLOAT	FLOAT
140	9	FEB-050Row:PBI-FT-A10:LoErrTrshR	ai	@PLC01/42 T=FLOAT	FLOAT
141	10	FEB-050Row:PBI-FT-A10:HiErrTrshR	ai	@PLC01/46 T=FLOAT	FLOAT
142	11	FEB-050Row:PBI-FT-A10:HiHiErrTrshR	ai	@PLC01/50 T=FLOAT	FLOAT
143	12	FEB-050Row:PBI-PLC-Line1:GapFlwHiHiErrTrshR	ai	@PLC01/54 T=FLOAT	FLOAT
144	13	FEB-050Row:PBI-PLC-Line1:GapFlwHiHiErrTrshR	ai	@PLC01/58 T=FLOAT	FLOAT
145	14	FEB-050Row:PBI-MV-A10:MismatchErrTimR	ai	@PLC01/62 T=SHORT	SHORT
	15	FEB-050Row:PBI-FT-A10:LoErrTimR	ai	@PLC01/64 T=SHORT	SHORT
	16	FEB-050Row:PBI-FT-A10:HiErrTimR	ai	@PLC01/66 T=SHORT	SHORT
	17	FEB-050Row:PBI-FT-A10:HiHiErrTimR	ai	@PLC01/68 T=SHORT	SHORT
	18	FEB-050Row:PBI-PLC-Line1:GapFlwHiHiErrTimR	ai	@PLC01/70 T=SHORT	SHORT
	19	FEB-050Row:PBI-PLC-Line1:GapFlwHiHiErrTimR	ai	@PLC01/72 T=SHORT	SHORT
	20	FEB-050Row:PBI-FCV-A10:KpR	ai	@PLC01/74 T=FLOAT	FLOAT
	21	FEB-050Row:PBI-FCV-A10:KpR	ai	@PLC01/76 T=FLOAT	FLOAT

BINARY ANALOG Configuration

Ess_nblm

TA EPICS ?

Ess_nblm

Click here to  Generate IOC Files
EPICS

AppName: Ess_nblm

Ioc directory path: D:\EPICS\S7PLCMODBUS\NBLM\UPDATE\AWL

PLC name: PLC01

PV Prefix:

Ip: 127.0.0.1

Top type: SARAF

Reading protocol Writing protocol

Protocol:  SIEMENS

Port: 2 000

InByte: 642

OutByte: 0

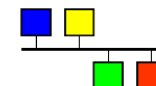
Endian type: BigEndian

Receive Timeout: 500

Send Interval: 100

BINARY ANALOG Configuration

EPICS



Data (D:) > EPICS > S7PLCMODBUS > NBLM > iocFiles	
Nom	Modifié le
deploy_ioc.sh	10/04/2019 14:57
IOC_MODBUS_CONFIGURATION.cmd	10/04/2019 14:57
IOC_S7PLC_CONFIGURATION.cmd	10/04/2019 14:57
iocEss_nblm.db	10/04/2019 14:57
iocEss_nblmInclude.dbd	10/04/2019 14:57
Makefile	10/04/2019 14:57

API dedicated EPICS :

- Based on CAJ (Channel Access for Java)
 - Pure Java implementation of Channel Access
 - <http://epics-jca.sourceforge.net/caj/>
- Read Process Variable
- Write Process Variable
- Run a Channel Access Server (Java IOC)

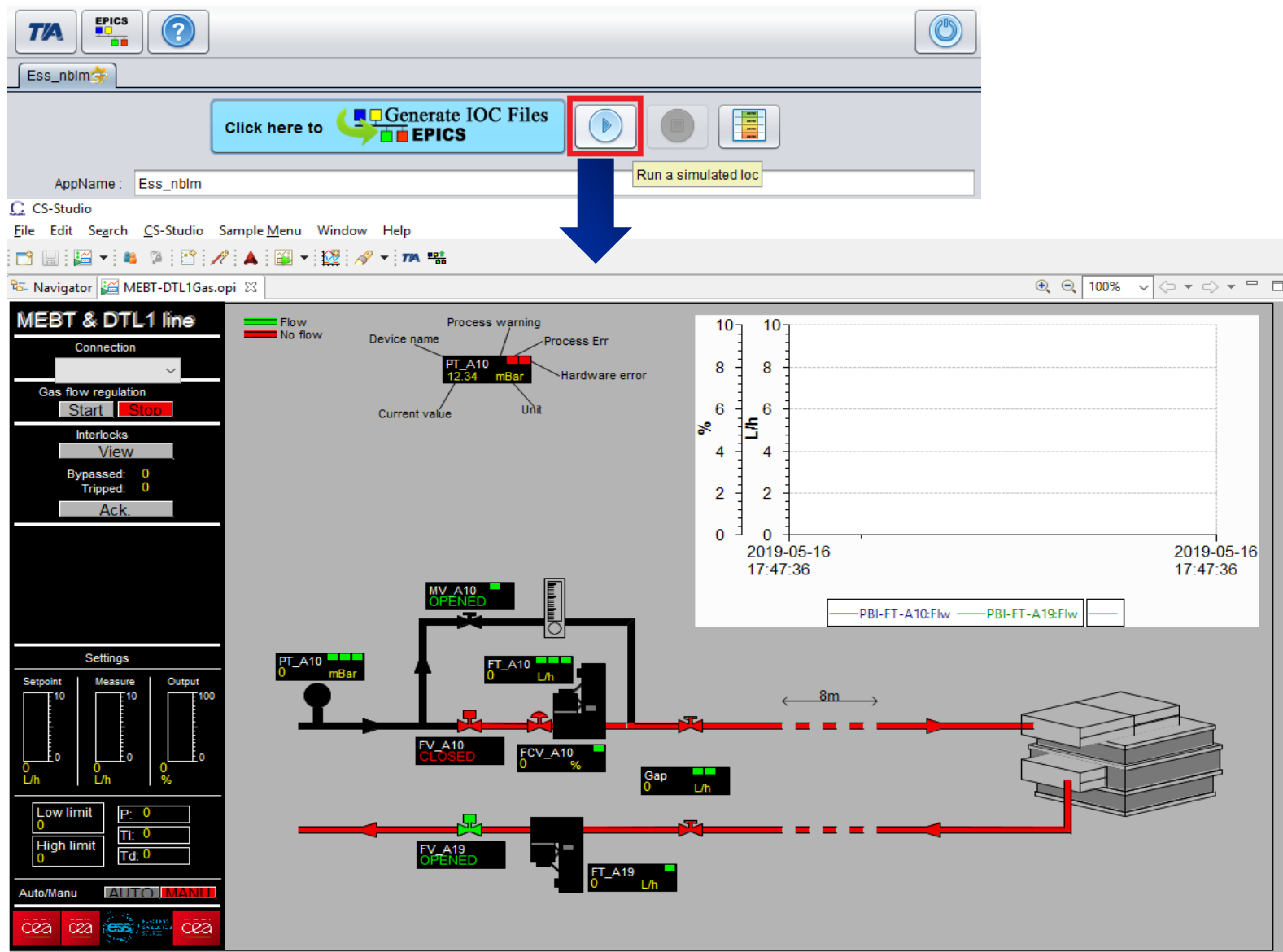


Applications :

- Simulated java IOC EPICS
- EPICS Generic Control Panel
- Muscade IOC EPICS
- ...



CAFE JAVA – RUN SIMULATED IOC



The screenshot displays the CAFE JAVA interface for running a simulated IOC. The top bar includes the TIA and EPICS logos, a help icon, and a button labeled "Click here to Generate IOC Files EPICS". Below this, the "AppName" field is set to "Ess_nblm". A red box highlights the "Run a simulated IOC" button (a play icon), with a large blue arrow pointing from it to the main simulation window.

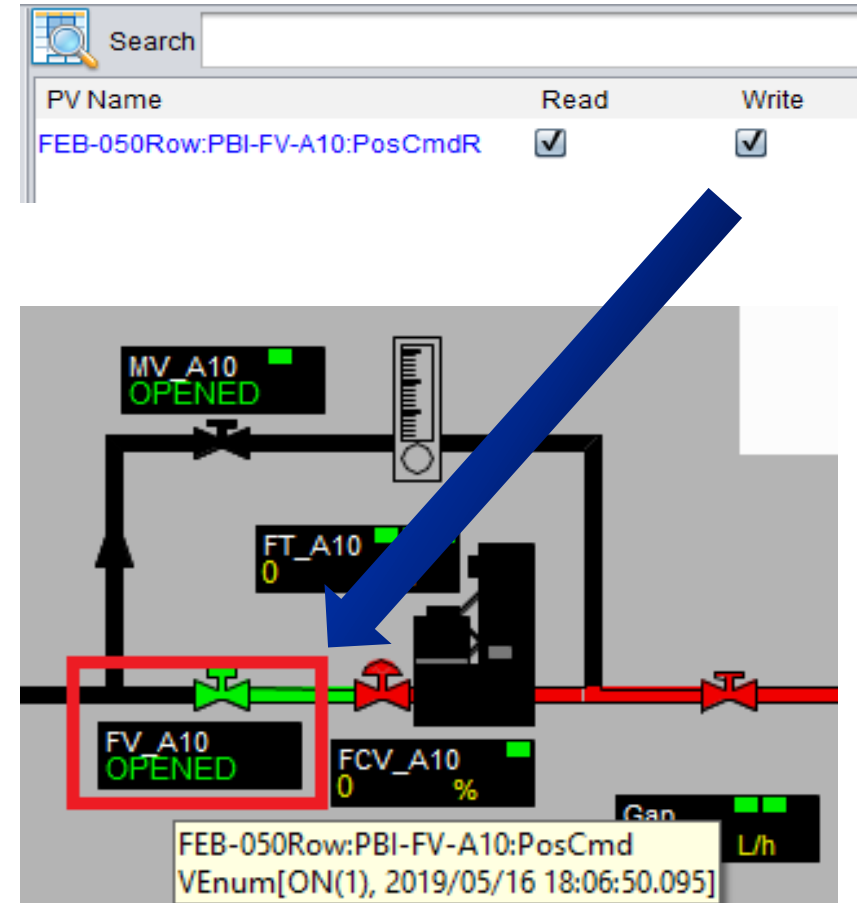
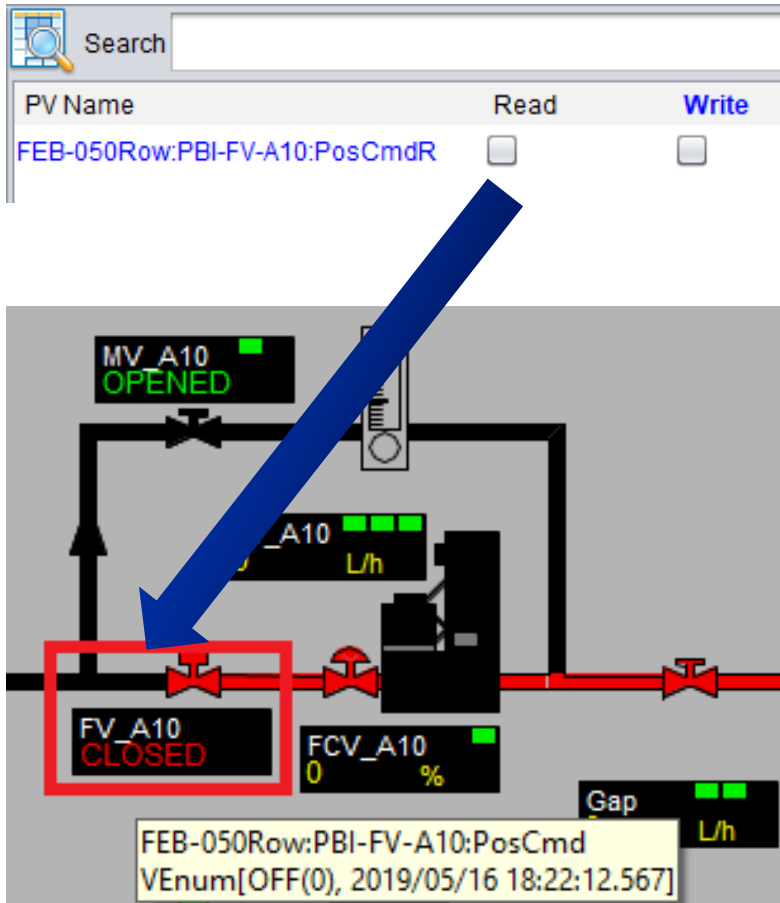
The main simulation window shows the "MEBT & DTL1 line" configuration. On the left, there are control panels for "Connection", "Gas flow regulation" (with Start and Stop buttons), "Interlocks" (with View button), and "Settings" (with Setpoint, Measure, and Output scales). The central area displays a process diagram with various components: PT_A10 (12.34 mBar), FT_A10 (0 L/h), FV_A10 (CLOSED), FCV_A10 (0 %), MV_A10 (OPENED), Gap (0 L/h), FV_A19 (OPENED), and FT_A19 (0 L/h). A legend indicates "Flow" (green line) and "No flow" (red line). A graph on the right shows two data series: "PBI-FT-A10:Flw" (blue line) and "PBI-FT-A19:Flw" (green line), both plotted against time from 2019-05-16 17:47:36 to 2019-05-16 17:47:36. The graph axes are labeled with values 0, 2, 4, 6, 8, 10, and units % and L/h. A distance of 8m is indicated between two components in the diagram.

- PV list built from EPICS db file
- Test the IOC without a synoptic
- Works in CSS or in stand alone mode

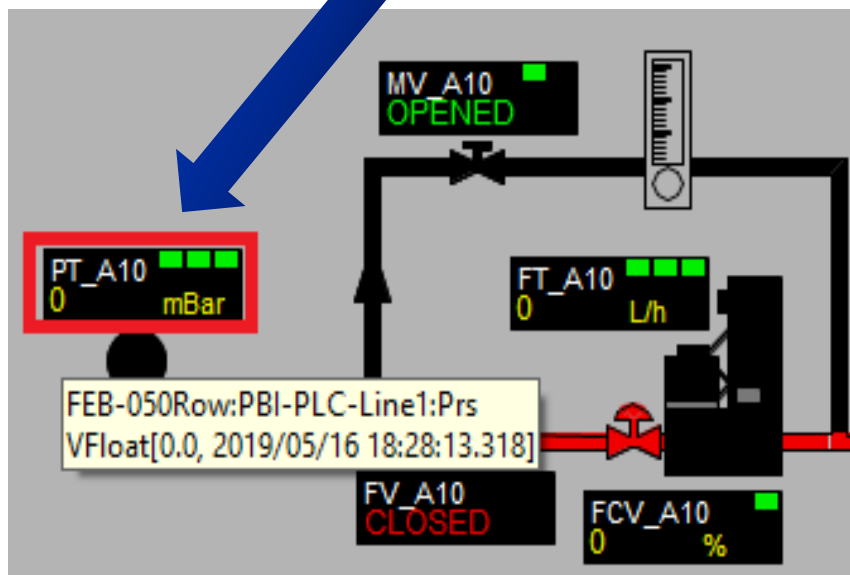


Data (D:) > EPICS > S7PLCMODBUS > NBLM > iocFiles		
Nom	Modifié le	
deploy_ioc.sh	10/04/2019 14:57	
IOC_MODBUS_CONFIGURATION.cmd	10/04/2019 14:57	
IOC_S7PLC_CONFIGURATION.cmd	10/04/2019 14:57	
iocEss_nblm.db	10/04/2019 14:57	
iocEss_nblmInclude.dbd	10/04/2019 14:57	
Makefile	10/04/2019 14:57	

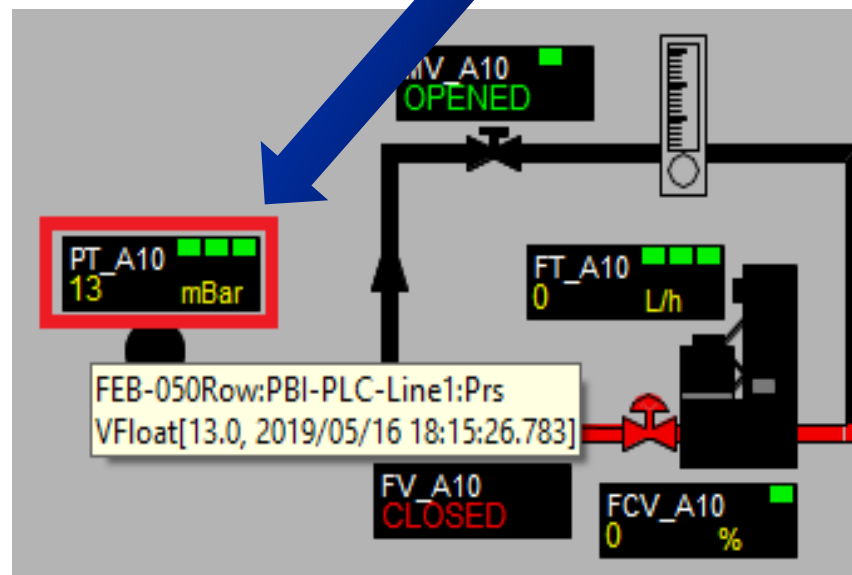
Ess_nblm
Search
Connection_status
FEB-050Row:PBI-FCV-A10:AutoManu
FEB-050Row:PBI-FCV-A10:AutoManuR
FEB-050Row:PBI-FCV-A10:HardErr
FEB-050Row:PBI-FCV-A10:HardErrR
FEB-050Row:PBI-FCV-A10:Kp
FEB-050Row:PBI-FCV-A10:KpR
FEB-050Row:PBI-FCV-A10:LimB
FEB-050Row:PBI-FCV-A10:LimBR
FEB-050Row:PBI-FCV-A10:LimH
FEB-050Row:PBI-FCV-A10:LimHR
FEB-050Row:PBI-FCV-A10:PosSP
FEB-050Row:PBI-FCV-A10:PosSPR
FEB-050Row:PBI-FCV-A10:Td
FEB-050Row:PBI-FCV-A10:TdR
FEB-050Row:PBI-FCV-A10:Ti
FEB-050Row:PBI-FCV-A10:TiR
FEB-050Row:PBI-FCV-A20:AutoManu
FEB-050Row:PBI-FCV-A20:AutoManuR
FEB-050Row:PBI-FCV-A20:HardErr
FEB-050Row:PBI-FCV-A20:HardErrR
FEB-050Row:PBI-FCV-A20:Kp
FEB-050Row:PBI-FCV-A20:KpR
FEB-050Row:PBI-FCV-A20:LimB
FEB-050Row:PBI-FCV-A20:LimBR
FEB-050Row:PBI-FCV-A20:LimH
FEB-050Row:PBI-FCV-A20:LimHR
FEB-050Row:PBI-FCV-A20:PosSP
FEB-050Row:PBI-FCV-A20:PosSPR
FEB-050Row:PBI-FCV-A20:Td
FEB-050Row:PBI-FCV-A20:TdR



PV Name	Read	Write
FEB-050Row:PBI-PLC-Line1:PrsR	0.0	0.0



PV Name	Read	Write
FEB-050Row:PBI-PLC-Line1:PrsR	13.0	13.0



MUSCADE
EMBEDDED SCADA

ARRAY 0: CR_VALREMP

TREND ARRAY HISTORICS REC

Previous Next Main

2019/05/17 14:23:06,044 +0200 C

CR_VALREMP	h1.CH1.TOTAL
0	0.00
DEPRESS	h1.CH2.FAILURE
0	0
E542A	h1.CH2.INSTANT
0.00	0.00
E542B	h1.CH2.TOTAL
0.00	0.00
E542C	h1.CH3.FAILURE
0.00	0
E542D	h1.CH3.INSTANT
0.00	0.00
ETAT_REMP	h1.CH3.TOTAL
0	0.00
S542A	
0.00	
S542B	
0.00	
S542C	
0.00	
S542D	
0.00	
h1.CH0.FAILURE	
0	
h1.CH0.INSTANT	
0.00	

muscade

Search

S542A
h1.CH0.FAILURE

Search

PV Name	Read	Write	LOLO
h1.CH0.FAILURE	☐	☐	

Search

PV Name	Read	Write	LOLO	LOW	HIGH	HIHI	DESC
S542A	0.0	0.0	-10.0	0	5.0	10.0	Muscade pv S542A

CS-Studio

File Edit Search CS-Studio Sample Menu Window Help

Display CENTER 100%

MUSCADE SERVER

S542A 0 Hauteur 0

S542A
VDouble[0.0, 2019/05/17 14:31:35.304]

h1.CH0.FAILURE OFF

h1.CH0.FAILURE
VEnum[OFF(0), 2019/05/17 14:31:35.304]

control system studio



ARRAY 0: CR_VALREMP

TREND ARRAY HISTORICS RE

Previous Next Main

2019/05/17 14:51:09,553 +0200

CR_VALREMP 0	h1.CH1.TOTAL 0.00
DEPRESS 0	h1.CH2.FAILURE 0
E542A 0.00	h1.CH2.INSTANT 0.00
E542B 0.00	h1.CH2.TOTAL 0.00
E542C 0.00	h1.CH3.FAILURE 0
E542D 0.00	h1.CH3.INSTANT 0.00
ETAT_REMP 0	h1.CH3.TOTAL 0.00
S542A 12.00	
S542B 0.00	
S542C 0.00	
S542D 0.00	
h1.CH0.FAILURE 0	

muscade



Search

S542A
h1.CH0.FAILURE

Search

PV Name	Read	Write	LOLO	LOW	HIGH	HIHI	DESC
S542A	12.0	12.0	-10.0	-5.0	5.0	10.0	Muscade pv S542A

CS-Studio

File Edit Search CS-Studio Sample Menu Window Help



Display

CENTER



100%



MUSCADE SERVER

S542A

12 Hauteur

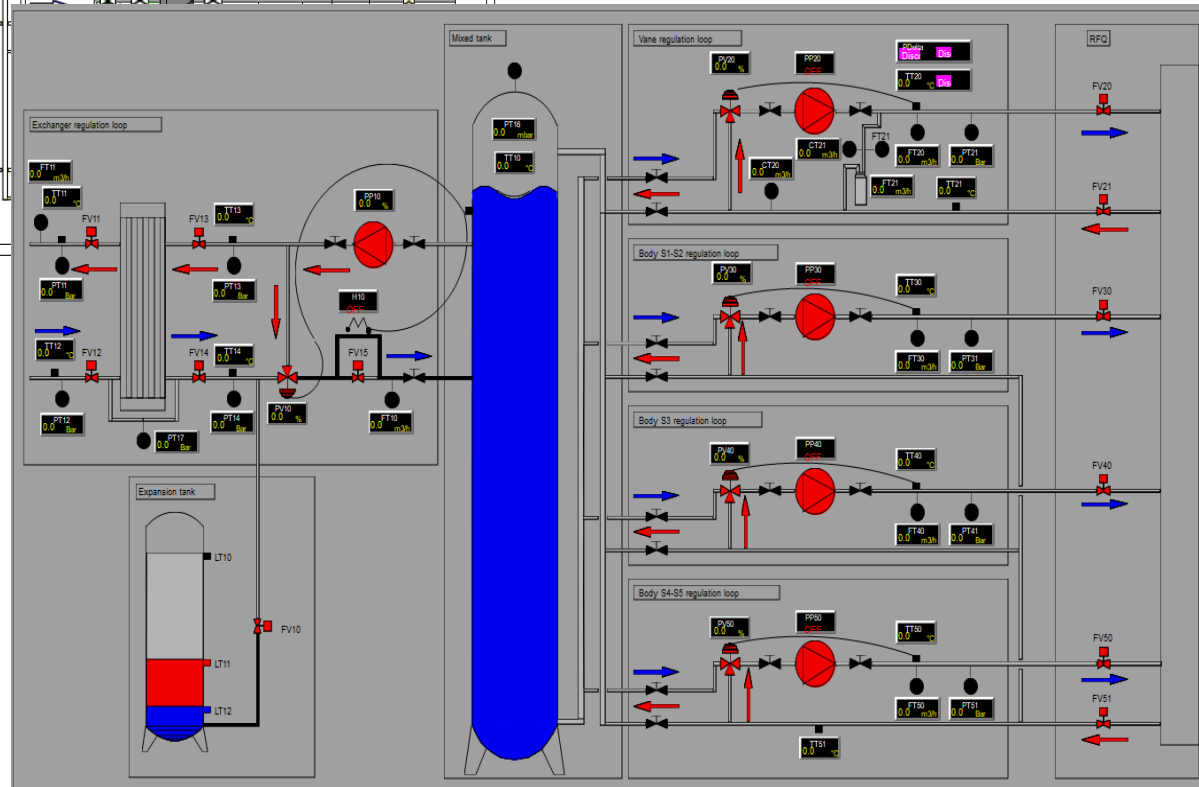
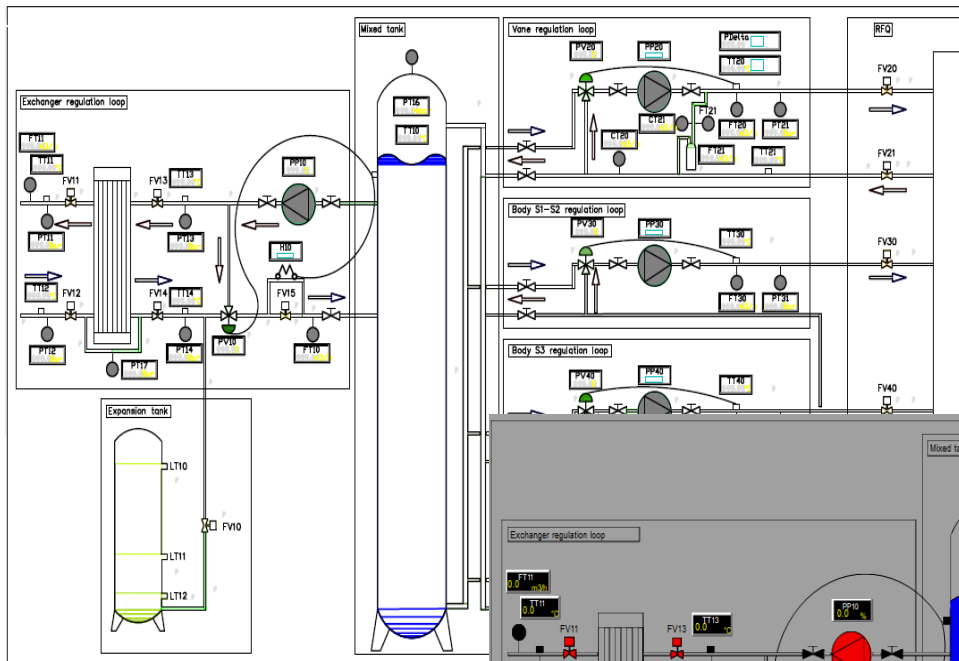
12

h1.CH0.FAILURE

OFF

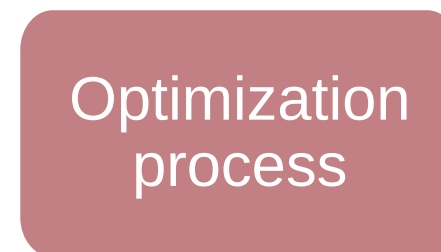
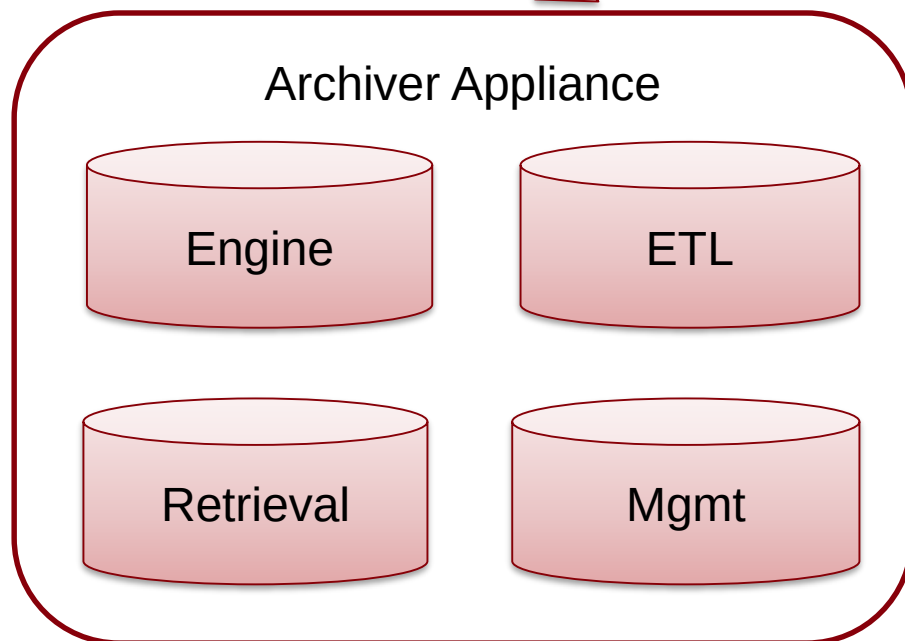
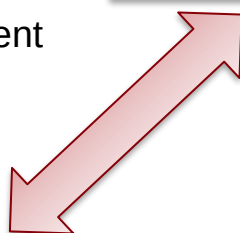
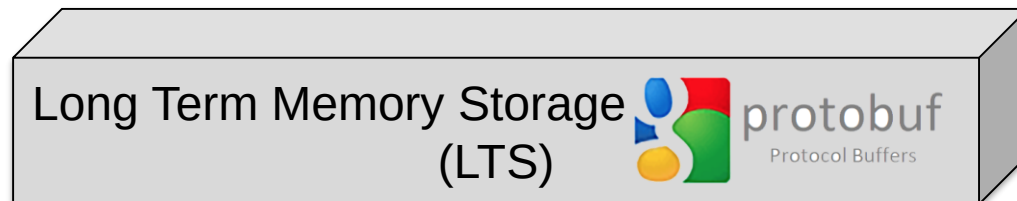
S542A
VDouble[12.0, MAJOR(HIHI_ALARM), 2019/05/17 14:49:14.460]





Small PC configuration: adapting Archive appliance

- Remove not interesting data
- Remove old data
- Not deployed for the moment



Memory Optimizer ON Archiver Appliance (MOONARCH) - Mozilla Firefox@irfulxis229

Memory Optimizer O... x

localhost:17669/moonarch/

MOON Archiver Appliance : Configuration page

User Configuration

Data depth: 1 S

Max Storage Duration: 5 year(s)

Optimization Mode: by File

PVs To Exclude: iphi:lebt:Thall
*BOB1_3_ON
ex : project.group:pvname ; can use wildcards (*)

Set Expert's Configuration

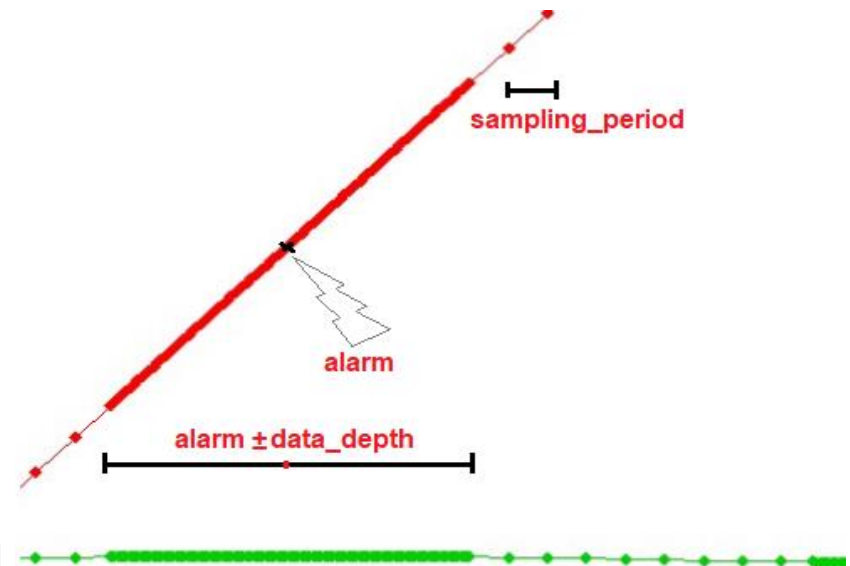
Load Default Configuration **Leave Changes** **Change Configuration**

Start Optimization

Stop Optimization

Parameters :

- Data depth
- Max storage duration
- Process variables to exclude
- Process period
- Optimization mode
- Process variables per task ...



Propriétés de : GBAR

Général Partage Sécurité Versions précédentes Personnaliser

GBAR

Type : Dossier de fichiers

Emplacement : D:\EPICS\ArchivingData

Taille : 1,32 Go (1 420 253 245 octets)

Taille sur le disque : 1,32 Go (1 420 967 936 octets)

Contenu : 394 Fichiers, 6 Dossiers



Memory size - 40 %

Propriétés de : GBAR

Général Partage Sécurité Versions précédentes Personnaliser

GBAR

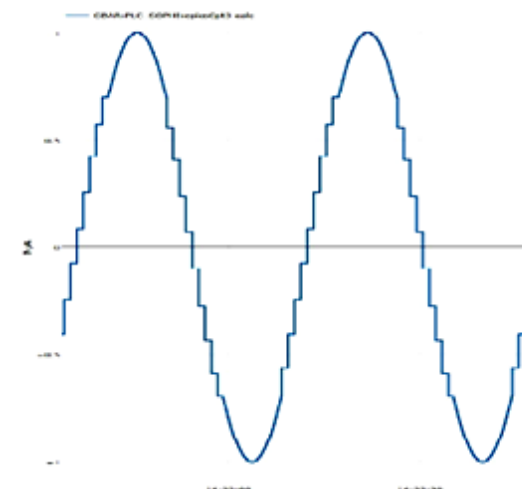
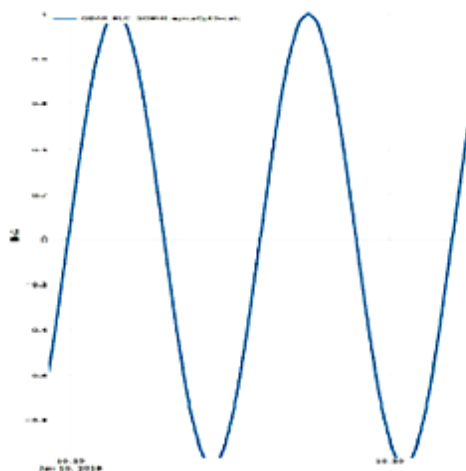
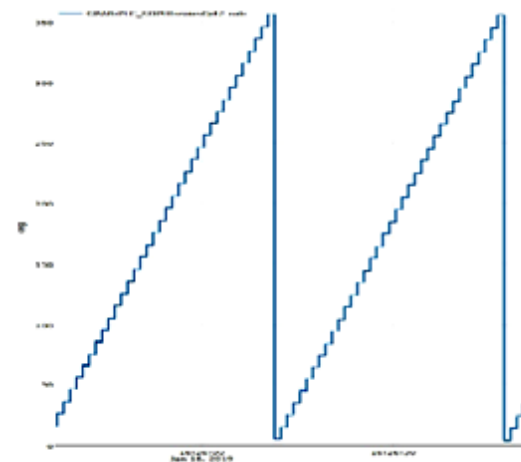
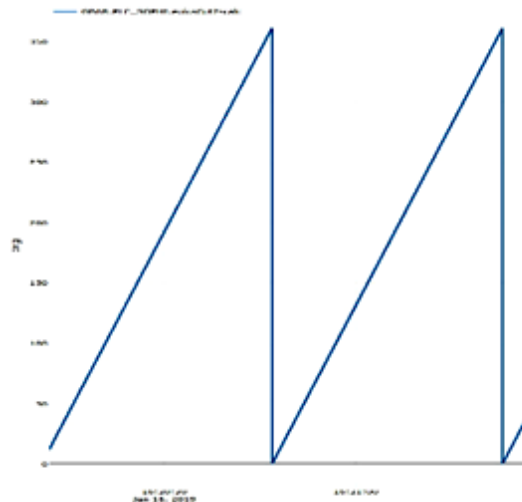
Type : Dossier de fichiers

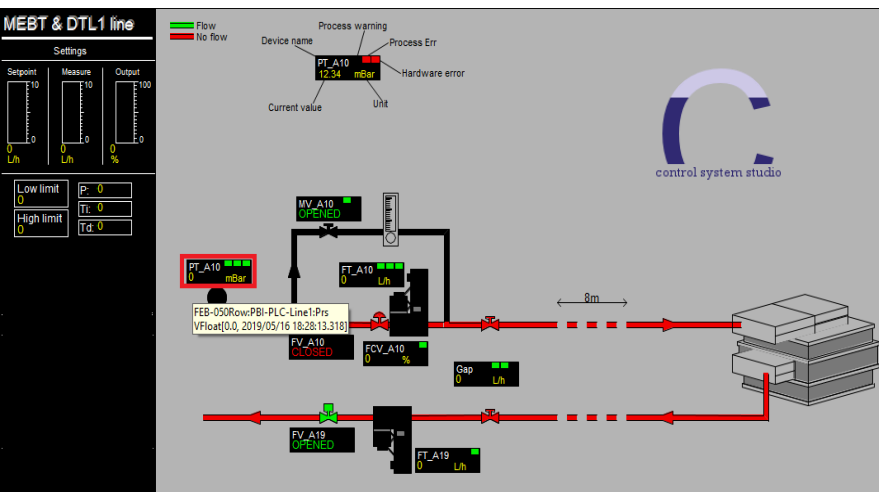
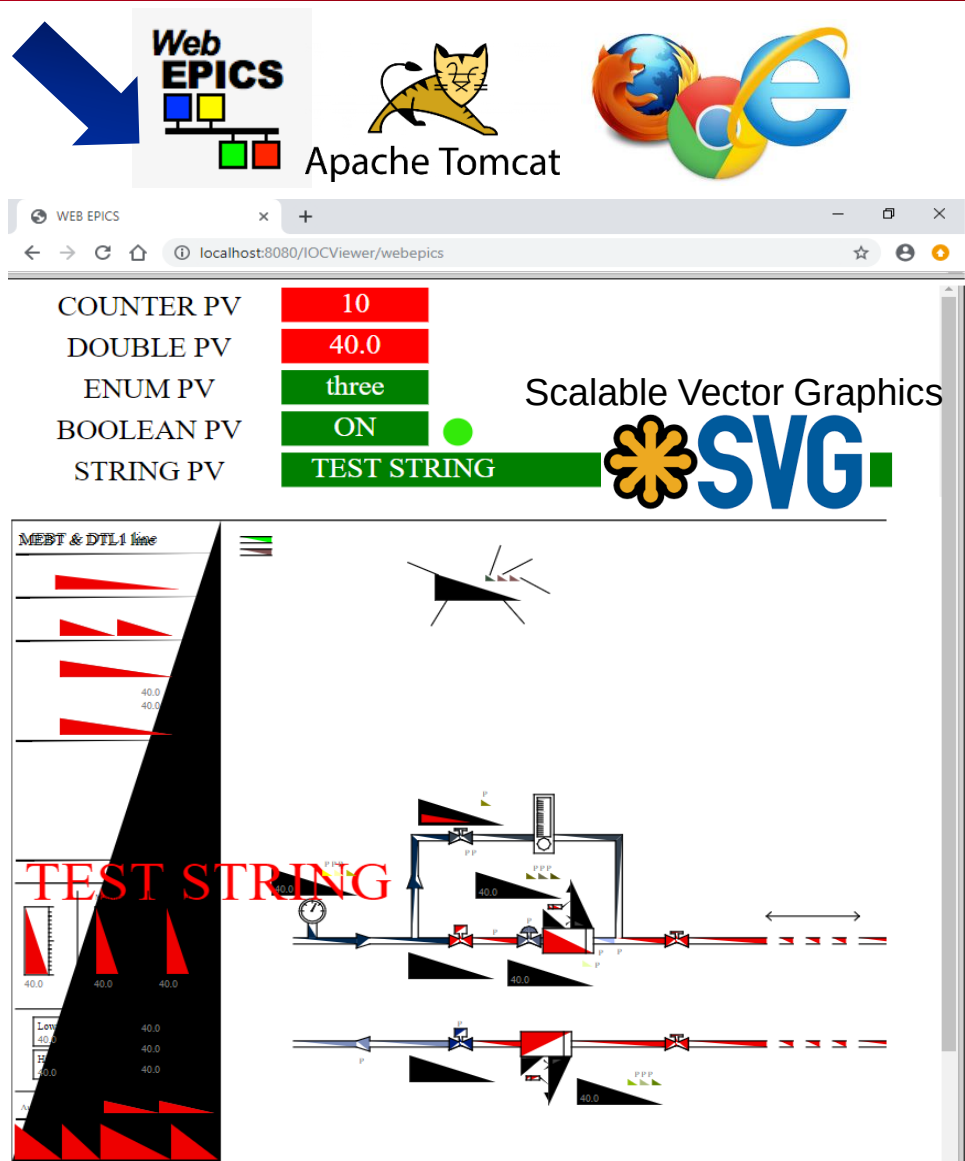
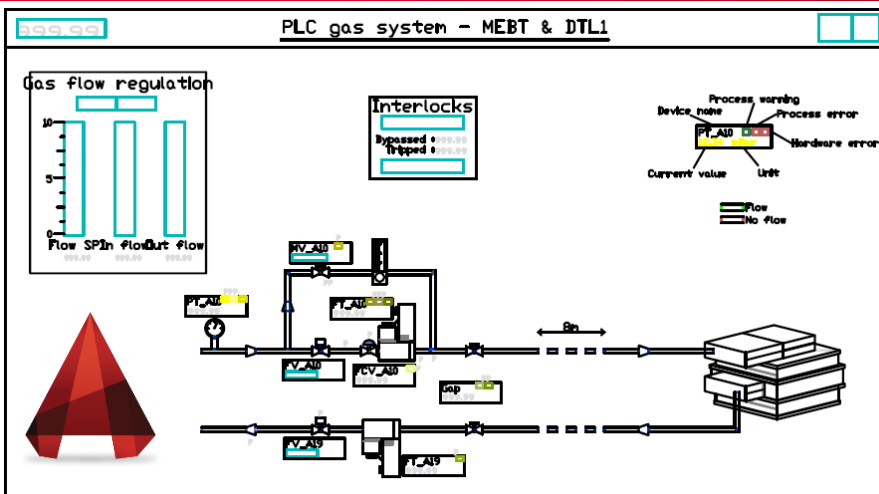
Emplacement : D:\EPICS\optimizedLts

Taille : 799 Mo (837 895 889 octets)

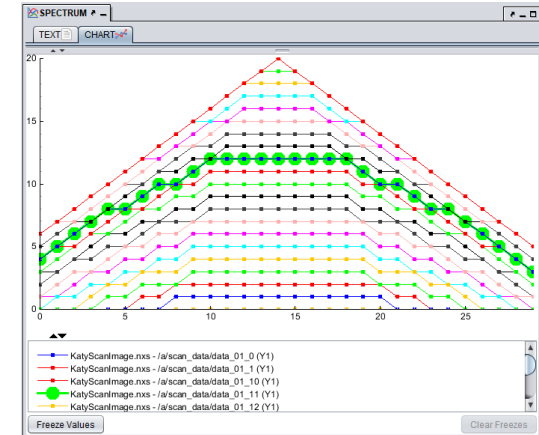
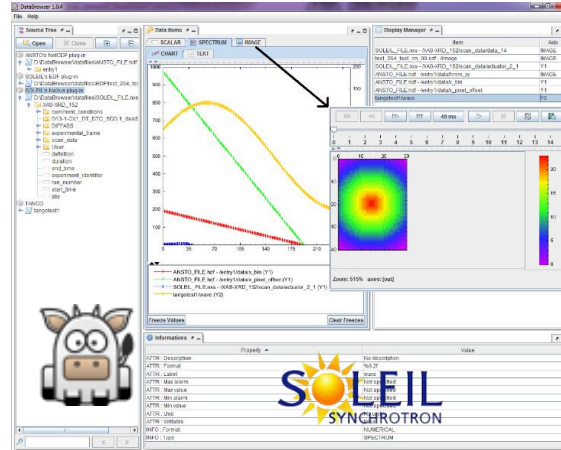
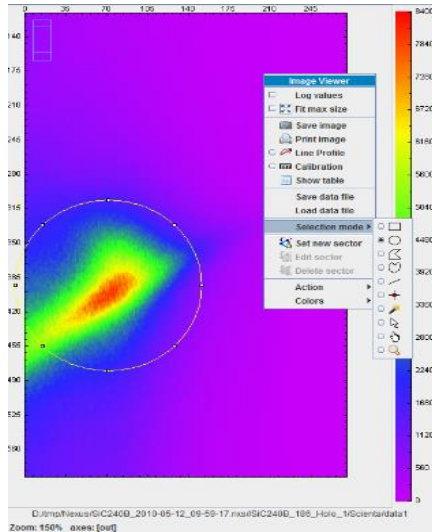
Taille sur le disque : 799 Mo (838 598 656 octets)

Contenu : 386 Fichiers, 6 Dossiers





<http://wiki.synchrotron-soleil.fr/tikiwiki/tiki-index.php?page=How-browse-downloaded-Nexus-files>



COMETE GUI



CDMA



Plugin



Plugin



Plugin

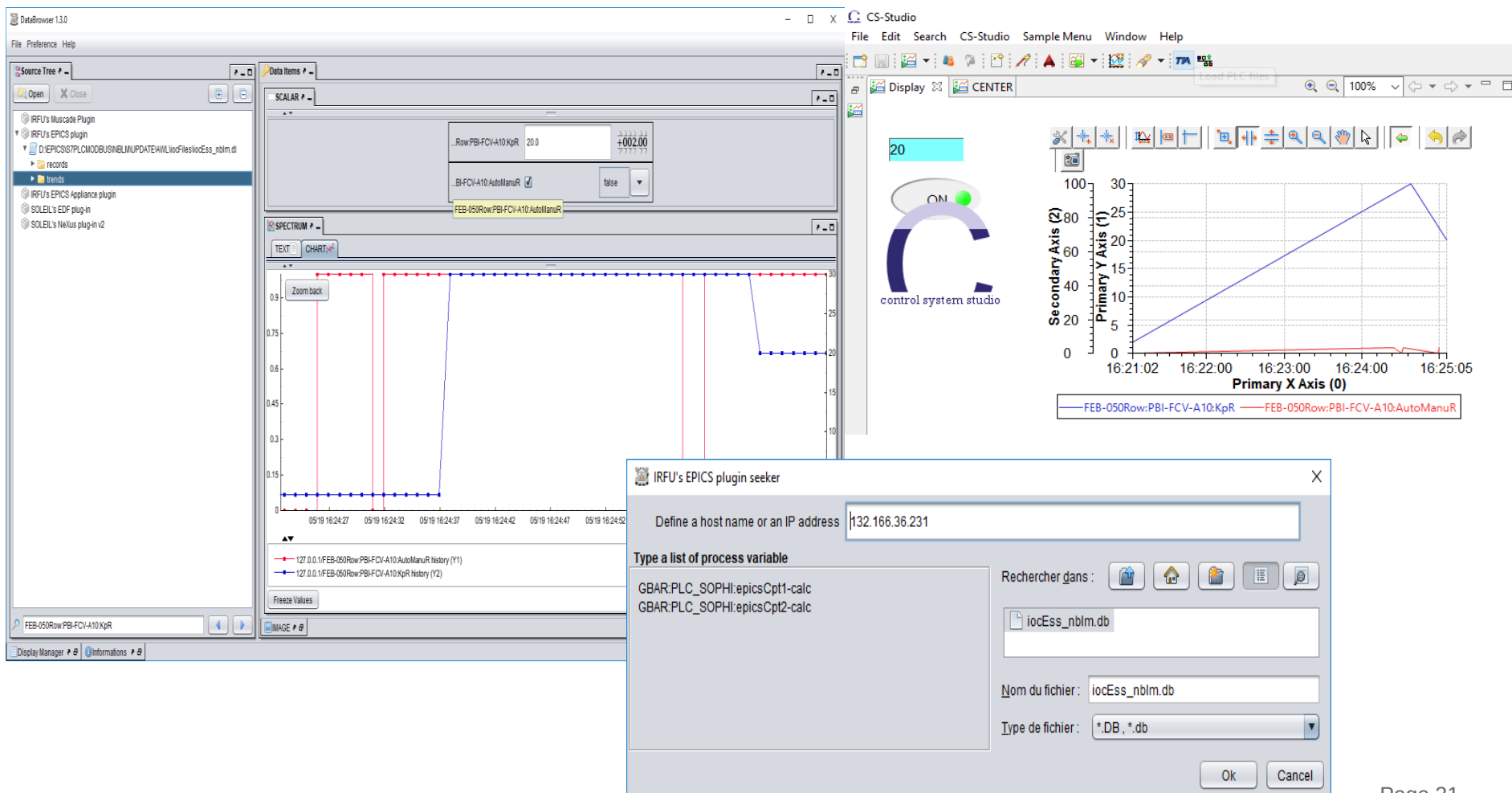
Plugin
EPICS

Plugin

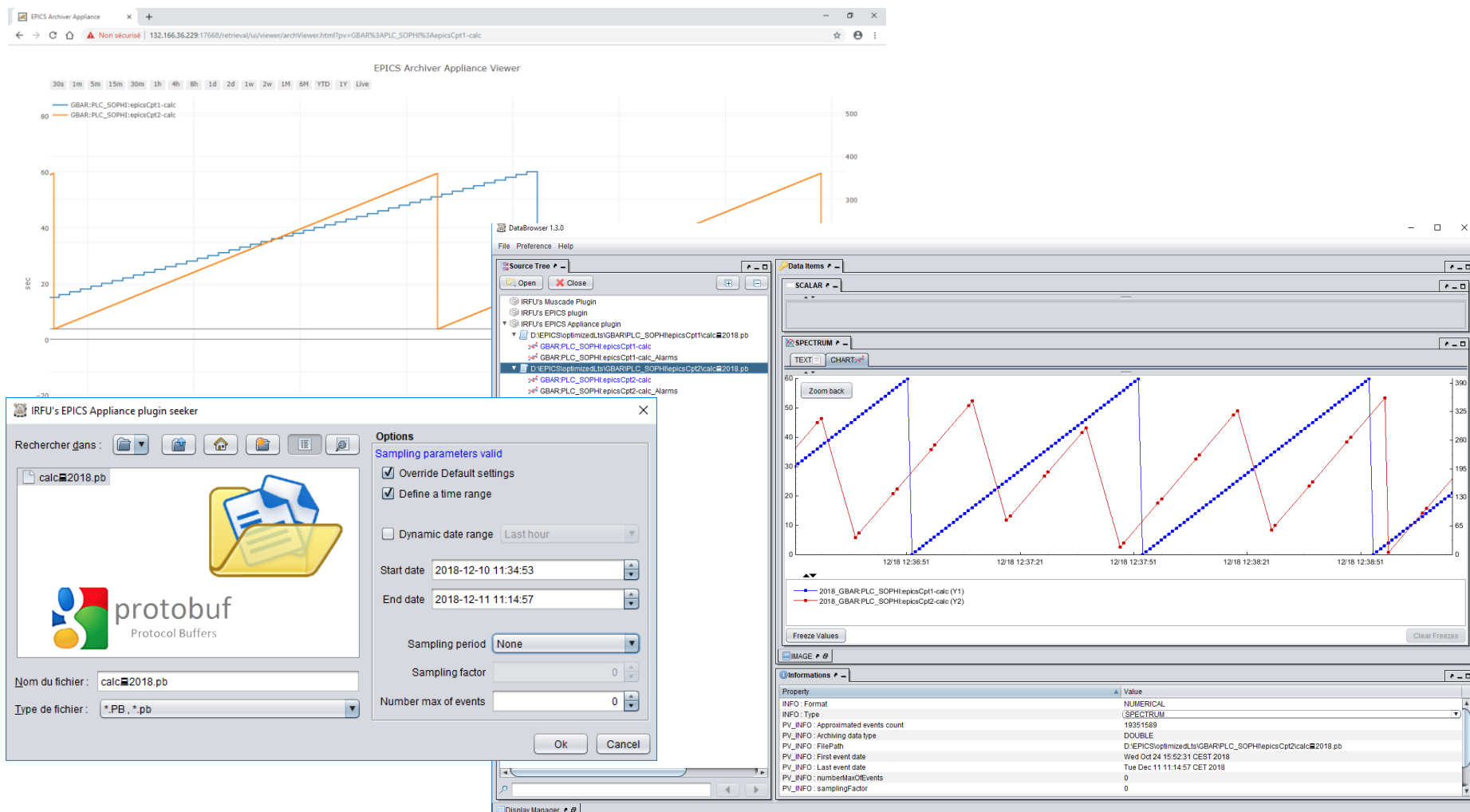


CAJ : Channel Access for Java to read & write PV

VisualDCT : EPICS Visual Database Configuration Tool for parsing a EPICS db file

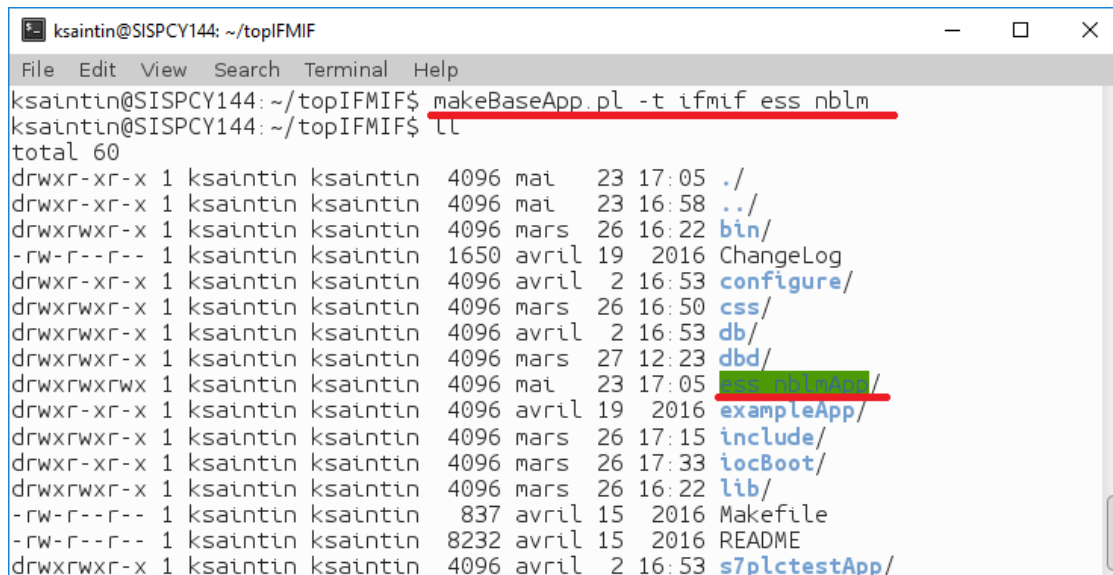
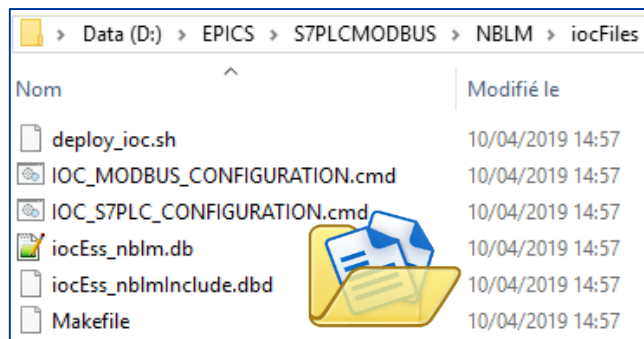


EPICS Archiver Appliance : Read Google Protobuf files



PLC Parser Tool :

- Generate script `deploy_ioc.sh` for IOC Files deployment in an EPICS's top folder



Synoptic visualization :

-  AutoCAD convert to BOB for PHEOBUS
-  AutoCAD convert to SVG for EPICS Web
-  mode, to replay a synoptic at a request time range



Deployment automation :

-  playbook for a full EPICS framework deployment on a small experiment



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